

RNN - (Recurrent Neural Network)

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RNN layman way Definition:

RNN is like a person reading a story one word at a time while remembering the previous words.

RNN is an AI model that remembers what it has already seen so it can better understand what comes next.

RNN is a type of neural network specifically designed to process sequential or time-dependent data.

Unlike a traditional neural network, an RNN has a memory that allows it to remember information from previous time steps while processing new inputs.

RNN Architectures

1. One-to-One (Single Input -> Single Output)

This is actually a standard neural network, not really an RNN

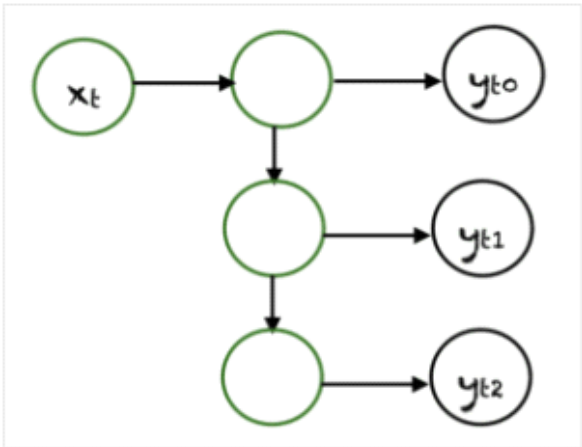


Use Case	Input	Output
Image Classification	Dog Image	Dog

Ex: Spam Detection, House Price Prediction, Disease Prediction, Credit Risk Prediction

2. One-to-Many (Single Input → Sequence Output)

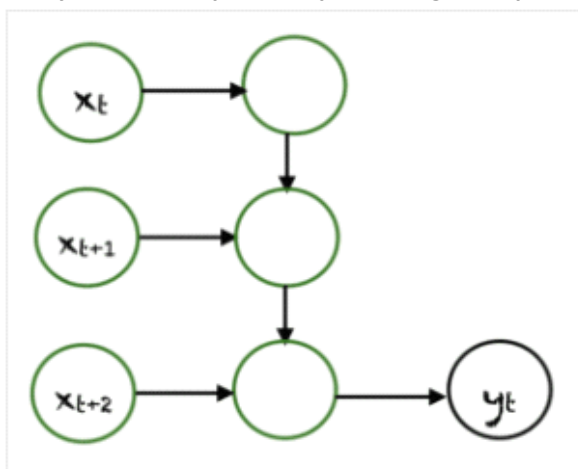
One input generates multiple outputs over time



Use Case	Input	Output
Image Captioning	(A boy playing football)	"A" "boy" "is"

		"playing" "football"
Music Generation	Genre = Classical	♪ ♪ ♪ ♪ ♪ ♪
Text Generation (ChatGPT-like decoder concept)	"Once upon a time"	Once upon a time there was a king...

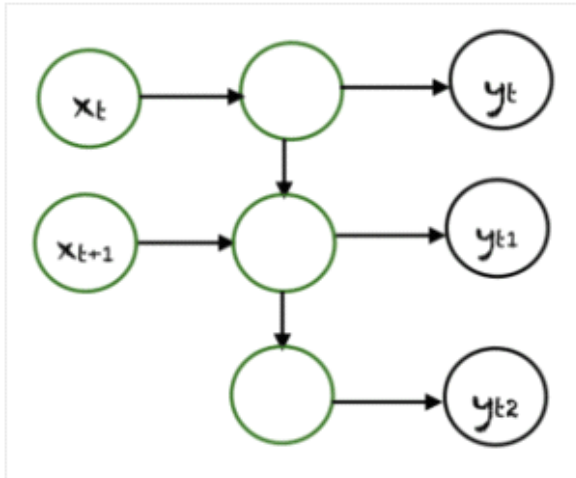
3. Many-to-One (Sequence Input → Single Output)



Multiple inputs arrive over time, but only one final prediction is made.

Use Case	Input	Output
Sentiment Analysis	"This movie is absolutely amazing"	Positive
Email Spam Detection	Congratulations! You won \$10,000. Click here.	Spam
Fake News Detection	Entire article	Fake / Real
Customer Review Classification	The delivery was late, but the product quality was excellent.	Satisfied

4. Many-to-Many (Sequence Input → Sequence Output)



Use Case	Input	Output	
Machine Translation	I love India	Je aime l'Inde	
Speech Recognition	Audio Frames ↓ Frame1 Frame2 Frame3 Frame4	Hello Everyone	
Video Activity Recognition	Frame1 Frame2 Frame3 Frame4	Walking Running Jumping Standing	
Named Entity Recognition (NER)	John lives in Bangalore	Word	Label
		John	Person
		Live	O
		In	O
		Bangalore	Location
Part-of-Speech (POS) Tagging	The cat runs	The Determiner cat Noun runs Verb	

Architecture	Input	Output	Real-Time Examples
One-to-One	Single	Single	Image Classification, House Price Prediction, Spam Detection
One-to-Many	Single	Sequence	Image Captioning, Music Generation, Story Generation, Text Generation
Many-to-One	Sequence	Single	Sentiment Analysis, Email Spam Detection, Fake News Detection, Review Classification
Many-to-Many	Sequence	Sequence	Machine Translation, Speech Recognition, Named Entity Recognition, POS Tagging, Video Frame Labeling